A quantum notion of ergodicity

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1. Motivation: quantisation of ergodic systems

The integrability-chaos perspective is a cornerstone of the understanding of the dynamics of classical systems. Chaotic dynamics, often thought of as "dynamics very sensitive to initial conditions", lies at the basis, on the one hand, of the phenomenology of many mechanical systems, and on the other hand it makes it possible to build the bridge between dynamics invariant under time reversal and the notions of statistical systems, in and out of equilibrium, as well as to the introduction of probability theory to define a deterministic system. Since the question "is this system chaotic?" can be asked in classical mechanics, it is natural to try to ask the same question for quantum systems.

Despite the existence of more or less canonical methods for quantising classical systems, it is important to acknowledge that the notions of integrability and chaoticity carry over into the quantum world in a highly non-trivial way. In the case of integrability it is clear from the start that it no longer makes sense to define an integrable system on the basis of the construction of a number of integrals of motion related to the dimension of the manifold, since the evolution of the system is given by a unitary transformation, so that all the operators that are simultaneously diagonalisable commute with that operator. A posteriori it was found that, to constrain a system through symmetries as happens for a classical integrable system, infinitely many conserved quantities must exist. In the case of the phenomenology of chaos another conceptual problem arises. Indeed, the motion is no longer characterised by initial conditions of position and velocity, but rather by the weight given to the eigenstates of the Hamiltonian, which describe a generic wave function and its evolution. What can one look for, then, for such eigenfunctions? Going back to the classical case, another way to picture a chaotic motion could be to think that a small ball of initial conditions will evolve in time uniformly over the whole space available to it, without the formation of small sets in which the points concentrate[1]. The mathematical formalisation of this idea is developed within ergodic theory, discussed in section 2. At the quantum level this condition seems simpler to transpose. One could ask whether, in a system that is classically chaotic, its quantum counterpart has eigenfunctions that are distributed uniformly over the space available to them. In section 3 we will see how to make this request precise, in the simplest case of a quantum system: the free particle. Restricting the treatment to the free particle, the need is clear to introduce an element of difficulty into the problem, which at least allows the presence of chaotic motions in the classical counterpart of the problem. This element will be given by the definition of a non-trivial geometry on which the motion is defined. Again in section 3 we will discuss which geometries lend themselves to defining chaotic motions both in the classical and in the quantum case. Finally, in section 4 some questions will be left unanswered, to try to highlight possible continuations of the following discussion.

[1] Obviously this heuristic, in its generality, is falsified by the existence of strange attractors.

2. Basics of ergodic theory

2.1 Measure-preserving transformations

In the most general setting, one can talk about ergodic theory given a sigma-finite measure space $(X, \mathcal{M}, \mu)$, and a transformation that preserves the measure.

$$\tau : X \to X, \quad \mu(\tau^{-1}(A)) = \mu(A), \quad \forall A \in \mathcal{M}. \tag{2.1}$$

A first simple observation is that, given an integrable function $f : X \to \mathbb{R}$, $f \circ \tau$ is integrable too, and in particular:

$$\int_X f(x)\ \mathrm{d}\mu(x) = \int_X f(\tau(x))\ \mathrm{d}\mu(x). \tag{2.2}$$

The choice of the set from which to take f $(L^1, L^2, ...)$ will be discussed later. The previous integrability property, carried over to the transformed functions $f \circ \tau$, can lead naturally to defining a mean, called the *ergodic mean*.

$$A(f) = \frac{1}{n} \sum_{k=0}^{n-1} f(\tau^k(x)). \tag{2.3}$$

In physics this formal apparatus often takes on a simple interpretation: X represents a space that describes the states (for example phase space in mechanics and statistical mechanics) on which a time-evolution function is defined, with discrete or continuous time $\tau^k(x)$ (acting on the point x and at time k). When this function represents the evolution of the system given by the Hamiltonian flow, Liouville's theorem requires that this transformation preserve the measure (or the mass) of the system.

Theorem 2.1: Liouville

Given a symplectic manifold (M, ω) and a Hamiltonian $H : M \to \mathbb{R}$, the Hamiltonian flow preserves the Liouville measure:

$$\mu_L(A) = \mu_L(\phi_H(A)), \quad \forall A \text{ measurable.}$$

Even though in this case the transport of equi-measurability goes in the opposite direction with respect to the definition given earlier for τ, the problem does not arise since the infinitesimal transformation is bijective. In this example, the function f plays the role of an observable on phase space that evolves in time (by composing it with the flow). At this point it is clear that the ergodic mean therefore represents a time average over the values to which an initial point has evolved. We observe that this kind of mean is easily reproducible experimentally, since it only requires pointwise measurements in time. This "experimental" property leads one to ask whether this kind of mean is related to global properties of a certain function/observable f, for example to the expected value or spatial mean: $\int_X f(x)\mathrm{d}\mu(x)$. For this to be true, the ergodic mean must lose its dependence on the initial position on which the time evolution is carried out. This kind of property turns out to be particularly natural when the system under study is chaotic.
A first theorem that holds in general for isometries on Hilbert spaces is the following.

Theorem 2.2: ergodic mean

Let $\mathcal{H}$ be a Hilbert space and T an isometry associated with it, and P the projector onto the subspace of vectors invariant under the isometry, i.e. such that $T(f) = f$. Then:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} T^k(f) = P(f) \quad \forall f \in \mathcal{H}$$

In the language used earlier one identifies $\mathcal{H} = L^2(X, \mathcal{M}, \mu)$ and $T \circ f(x) = f(\tau(x))$. Note that the limit written in the theorem above is to be understood in norm, induced by the Hilbert space structure. If one requires $f \in L^1$, through the use of maximal functions it is possible to find a pointwise counterpart to the theorem above.

Theorem 2.3: pointwise ergodic mean

Given a finite measure space with $\mu(X) = 1$, and an $f \in L^1(X, \mathcal{M}, \mu)$, then for a.e. $x \in X$ the limit exists:

$$P'(f) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} (f(\tau^k(x)))$$

Moreover $P'(f) = P(f)$ if $f \in L^2(X)$.

2.2 Ergodic transformations

So far we have not yet mentioned how to link the ergodic mean with the spatial one. In general, a corollary of theorem (2.1) is the following:

Theorem 2.4

Under the hypotheses of theorem (2.1) one obtains:

$$\int_X |P'(f(x))| \mathrm{d}\mu(x) \leq \int_X |f(x)| \mathrm{d}\mu(x)$$

To be able to improve the theorem, it is necessary to add a hypothesis on the transformation $\tau(x)$. We call a measure-preserving transformation *ergodic* if, given a set $A \in \mathcal{M}$ such that $\tau(A) = A$, then $\mu(A) = 0$ or 1. Equivalently, a transformation can be defined to be ergodic by requiring that the only functions invariant under τ, i.e. such that $f \circ \tau = f$, are the constant functions. With this condition it is possible to prove the hoped-for theorem.

Theorem 2.5

Let f be an integrable function of X and τ an ergodic transformation; then:

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} (f(\tau^k(x))) = \int_X f \mathrm{d}\mu, \quad \text{for } a.e.\ x \in X.$$

As anticipated, we observe that the RHS has no dependence on the value of x, on which a priori the ergodic mean could depend. Even without providing a proof, it is possible to give a

justification of the theorem by observing that if a transformation is ergodic and f is in $L^2(X)$, $P'(f) = P(f) = (f, 1)1$, while if it is in $L^1(X)$, one can write $f = g + h, g \in L^2(X), \|P'(h)\|_{L^1} \leq \|h\|_{L^1} < \epsilon$, so that:

$$\left|P'(f) - \int f \mathrm{d}\mu\right| = \left|P'(h) + \int (g - f)\mathrm{d}\mu\right| \leq \left\|P'(h)\right\|_{L^1} + \|h\|_{L^1} \leq 2\epsilon. \tag{2.4}$$

A notion that implies ergodicity is the so-called *mixing* condition, for which, $\forall E, F \in \mathcal{M}$, $\mu(\tau^{-n}(E) \cap F) \to \mu(E)\mu(F)$ as $n \to \infty$. The implication is obtained by observing that the mixing property in terms of functions can be written as $(T^n(f), g) \to (1, f)(1, g)$. At this point it follows directly for $L^2(X)$ functions:

$$(P(f), g) = \lim_{n\to\infty} \frac{1}{n}(\sum_{k=0}^{n-1} T^k(f), g) = \lim_{n\to\infty} \frac{1}{n} \sum_{k=0}^{n-1} (T^k(f), g) = (f, 1)(1, g).$$

But if g is orthogonal to the subspace of constant functions the scalar product is zero, which is equivalent to saying that $P(f)$ belongs to the subspace of constant functions. We conclude the section by reporting some simple examples of measure-preserving, ergodic and mixing transformations.

1. on the circle written as $X = \mathbb{R}/\mathbb{Z}$ and having as measure the one induced by the quotient from the Lebesgue measure, the translations/rotations $x \to x + \alpha$ preserve the measure. In the case in which $\alpha \in \mathbb{R}/\mathbb{Q}$ one can then show that the transformation is ergodic.

2. considering again $X = \mathbb{R}/\mathbb{Z}$, the transformation $f(x) = 2x$ is mixing. The fact that it preserves the measure is found by observing that the preimage of every connected set is made of two non-intersecting sets with measure equal to half of the original set. The mixing property instead can be guessed by checking the equivalent property on the basis e^{ikx}.

3. Quantum ergodic theorem

3.1 Ergodicity on Riemannian manifolds

Before stating the quantum ergodicity theorem, it is useful to see how what was defined in the previous chapter applies to the context of Riemannian manifolds. Let (M, g) be a Riemannian manifold; for every pair (p, v) with $p \in M$ and $v \in T_pM$ it is possible to define uniquely a geodesic flow $\phi^t : M \to M, \quad \phi^{t_0}(x) = x, \quad \frac{\mathrm{d}}{\mathrm{d}t}\phi^{t_0}(x) = v$ More generally, this flow is defined on the tangent bundle of the manifold

$$\phi^t : TM \to TM$$

. This flow is called geodesic since it satisfies the geodesic equation:

$$\nabla_{\dot\gamma}\dot\gamma = 0. \tag{3.1}$$

Observing that:

$$\frac{\mathrm{d}}{\mathrm{d}t} g(\dot\gamma, \dot\gamma) = 2g(\nabla_{\dot\gamma}\dot\gamma, \dot\gamma) = 0. \tag{3.2}$$

The geodesic flow can be restricted to the sphere bundle.

$$\phi^t : SM \to SM, \qquad SM = \{(x, v) \in TM, \|v\| = 1\}$$

. Looking for natural measures, the metric defines a measure on the manifold $dvol_g(x) = \sqrt{|\det(g)|}\mathrm{d}x_1...\mathrm{d}x_n$, while on the cotangent bundle T^*M there is always the Liouville measure, which in canonical coordinates is written $\mathrm{d}\mu_L(x) = \mathrm{d}x_1...\mathrm{d}x_n\mathrm{d}p_1\mathrm{d}p_n$ We observe that through the musical isomorphism one can also define, by push-forward, a measure on the tangent bundle. On the cosphere and sphere bundles too it is possible to define a canonical measure, by defining the contact 1-form $\alpha = \theta_{|\Sigma}$ from the restriction of the Liouville 1-form, and considering the $2n-1$ form $\alpha \wedge (\mathrm{d}\alpha)^{n-1}$. Note that, from the theory of disintegration of measures, this can also be written as $\mathrm{d}vol_g(x)\mathrm{d}S_x(\xi)$. Having defined the measure, the ergodic theorem can be formulated as follows.

Theorem 3.1: Ergodic theorem, Riemannian manifold

Let (M, g) be a compact Riemannian manifold and let ϕ^t be the geodesic flow on the cosphere bundle S^*M, which preserves the Liouville measure

$$\mu = \alpha \wedge (\mathrm{d}\alpha)^{n-1}.$$

Suppose that the flow ϕ^t is ergodic with respect to μ. Then for every function $f \in L^1(S^*M, \mu)$ and for μ-almost every $\xi \in S^*M$

$$\lim_{T\to\infty} \frac{1}{T} \int_0^T f(\phi^t(\xi))\, dt = \int_{S^*M} f\, d\mu.$$

We conclude the section with a last link between symplectic and Riemannian geometry. It is possible to show that the geodesic flow defined above is equivalent to the Hamiltonian flow corresponding to the free particle:

$$H = \frac{1}{2} g_{ij}(x) p^i p^j \tag{3.3}$$

Where the p_i are the canonical coordinates, the vector elements of the cotangent bundle (x, p). This equivalence can be found by writing the Hamilton-Jacobi equations, which imply the geodesic equations associated with the metric.

3.2 Quantum ergodicity

Let us now move to the "quantum" formulation of the ergodicity problem. Given a compact Riemannian manifold (M, g) without boundary, one defines the eigenvalue problem:

$$-\nabla_g^2 u_j = \lambda_j u_j. \tag{3.4}$$

Where the eigenvalues $\lambda_j \geq 0$ and j is chosen so that $\lambda_j \leq \lambda_{j+1}$. $\nabla_g^2 : L^2(M) \to L^2(M)$ is the Laplacian defined on the manifold, which in coordinates can be written as:

$$\nabla^2 f = \frac{1}{\sqrt{|g|}}\, \partial_i\Big(\sqrt{|g|}\, g^{ij}\, \partial_j f\Big) \tag{3.5}$$

From now on we will say that a certain sequence u_{j_k} is *equidistributed in physical space* if the measure $|u_{j_k}|^2 dvol_g(x)$ converges weakly to the normalised volume measure, i.e. $\forall a \in C^\infty(M)$:

$$\int_M a(x)|u_{j_k}|^2 \mathrm{d}vol_g(x) \to \frac{1}{vol_g(M)} \int_M a(x)\mathrm{d}vol_g(x), \quad as\ k \to \infty. \tag{3.6}$$

Note that for the formula written above to make sense the manifold must have finite volume. Now we want to find a correspondence between classical Hamiltonian physics, characterised by the use of functions on the cotangent bundle of a manifold, and quantum mechanics, which deals with operators on the space of functions $L(M)$ on the manifold. Let $a(x,\xi) \in C_c^\infty(T^*M)$ be a classical observable; we define the associated *quantum observable*:

$$Op_h(a) : L^2(M) \to L^2(M), \quad Op_h(a)f(x) = (2\pi h)^{-n} \int_{\mathcal{R}^n} e^{\frac{i(x-y)\xi}{h}} a\Big(\frac{x+y}{2}, \xi\Big) f(y)\mathrm{d}y\mathrm{d}\xi. \tag{3.7}$$

h is also called the *semiclassical parameter*, with the idea that it imitates the role of Planck's constant, and that in the limit $h \to 0$ the identities of the classical functions on the cotangent space are recovered.

Theorem 3.2: semiclassical limits

For every $a, b \in C_c^\infty(T^*M)$:

- $Op_h(a)Op_h(b) = Op_h(ab) + O(h)_{L^2}$;
- $Op_h(a)^* = Op_h(\bar{a}) + O(h)_{L^2}$

To have a semiclassical parameter in the eigenvalue problem defined above, one defines $h_j = \frac{1}{\lambda_j}$, so as to obtain the rewriting:

$$(-h_j^2 \nabla_g^2 - 1)u_j = 0 \tag{3.8}$$

We observe that the transform Op behaves well under integration of purely spatial functions $a \in C^\infty(M)$:

$$\int_M a(x)|u_j(x)|^2 \mathrm{d}vol_g(x) = \langle Op_{h_j}(a)u_j, u_j\rangle. \tag{3.9}$$

Where the scalar product is that of $L^2(M, \mathrm{d}vol_g)$. A notion that generalises equidistribution in physical space is *equidistribution in phase space*, in which one requires of a subsequence u_{j_k} that $\forall a \in C_c^\infty(T^*M)$:

$$\langle Op_{h_{j_k}}(a)u_{j_k}, u_{j_k}\rangle \to \int_{S^*M} a(x,\xi)\mathrm{d}\mu_L(x,\xi). \tag{3.10}$$

Where S^*M is the cosphere bundle and μ_L the Liouville measure normalised to 1 associated with it. It is clear that equidistribution in physical space is implied by that in phase space. The reason why one restricts to the cosphere bundle comes from the fact that in the semiclassical limit ($h_j \to 0$) the eigenfunctions live in that space. The following lemma is indeed easy to show.

Lemma 3.3

Let $a \in C_c^\infty(T^*M), \quad a|_{S^*M} = 0 \implies \Big\|Op_{h_j}(a)u_j\Big\|_{L^2} = O(h_j)$.

The lemma follows from the semiclassical result $-h^2\nabla_g^2 - 1 = Op_h(|\xi|_g^2) - 1 + \ O(h)$ and from the decomposition $Op_h(a) = Op_h(b)(-h^2\nabla_g^2 - 1) + \ O(h)$. The ergodicity condition of the classical geodesic flow defines some cases in which the eigenvectors satisfy the equidistribution properties. The formalisation of this idea is given by the quantum ergodic theorems. We now state three theorems. The first is implied by the second (trivially) and the second is implied by the third. In the following section the proof of the third theorem will be given, together with the implication of the second.

Theorem 3.4: Shnirelman

Let (M, g) be a compact Riemannian manifold without boundary. Let the geodesic flow be ergodic with respect to the Liouville measure. Then there exists a density-1 subsequence of eigenfunctions that equidistributes in physical space.

Where a subsequence is said to have density 1 if $\frac{\#\{k|\lambda_{j_k} \leq R\}}{\#\{j|\lambda_j \leq R\}} \to 1$ as $R \to \infty$.

Theorem 3.5: Quantum ergodicity

Let (M, g) be a compact Riemannian manifold without boundary. Let the geodesic flow be ergodic with respect to the Liouville measure. Then there exists a density-1 subsequence of eigenfunctions that equidistributes in phase space.

Theorem 3.6: Quantum ergodicity, integrated form

Let (M, g) be a compact Riemannian manifold without boundary. Let the geodesic flow be ergodic with respect to the Liouville measure. Consider a generic $a \in C_c^\infty(T^*M)$ and let:

$$L_a := \int_{S^*M} a \mathrm{d}\mu_L, \quad V_j(a) := \langle Op_{h_j}(a)u_j, u_j \rangle.$$

Then:

$$R^{-n} \sum_{\lambda_j \in [R,2R]} |V_j(a) - L_a|^2 \to 0 \quad as\ R \to \infty$$

3.3 Proof

The proof of these theorems is organised in several steps. First we will show how theorem 3.6 implies theorem 3.5, and then the proof of theorem 3.6 will be shown. The first lemma to prove estimates the number of eigenvalues smaller than a certain value R.

Lemma 3.7: Weyl's law

$$\#\{j|\lambda_j \leq R\} = \frac{\omega_n}{(2\pi)^n} \mathrm{vol}_g(M) R^n + \ O(R^n).$$

Where $n = dim(M)$ and ω_n is the volume of the unit n-sphere. The lemma can be proved through two results of semiclassical quantisation:

- One defines a *spectral filter* on real functions $\chi \in C_c^\infty(\mathcal{R})$ and then on the Laplacian operator through its application on the basis of eigenfunctions$\chi(-h^2\nabla_g^2)u_j = \chi(-h^2\lambda_j^2)u_j$ + linearity. Equivalently one shows that $\chi(-h^2\nabla_g^2) = Op_h(a_\chi)$, with $a_\chi(x,\xi,h) = \chi(|\xi|_g^2) + O(h)$.

- Let a be a function on the cotangent bundle; then the operator $Op_h(a)$ satisfies:

$$tr(Op_h(a)) = (2\pi h)^{-n}(\int_{T^*M} a(x,\xi)\mathrm{d}\xi\mathrm{d}x) + O(h)$$

  .

  Where the error comes only from curved-spacetime effects. The formula is easily obtained once the relation $tr(Op_h(a)) = \int \mathrm{d}x K_a(x,x)$ is found, where $K_a(x,y) = (2\pi h)^{-n}\int e^{\frac{i(x-y)\xi}{h}} a(\frac{x+y}{2})\mathrm{d}\xi$, which in turn is obtained from the definition + the identity $\sum_n \bar{e}_n(x)e_n(y) = \delta(x-y)$.

Combining these two facts one obtains:

$$\sum_j \chi((h\lambda_j)^2) = tr\chi(-h^2\nabla_g^2) = (2\pi h)^{-n}(\int_{T^*M} \chi(|\xi|_g^2)\mathrm{d}\xi\mathrm{d}x + \ O(h)).6$$

The claim is obtained by requiring that χ be a smooth function close to the indicator function of $[0,1]$.

Let us now see how to show that theorem 3 implies theorem 2. the claim of theorem 3 requires that:

$$R^{-n} \sum_{\lambda_j \in [R,2R]} |V_j(a) - L_a|^2 \to 0 \quad as\ R \to \infty$$

Knowing from Weyl's law that the number of eigenvalues is of size R^n, the previous statement says that the mean of $|V_j(a) - L_a|^2$ (multiplied by a constant) tends to 0 for $R \to \infty$. For probabilistic reasons, then, the number of j such that $|V_j(a) - L_a|^2 \geq \epsilon(R)$, with $\epsilon(R) \to 0$, cannot be of size larger than $\epsilon(R)$. Considering R sufficiently large (such as $R = 2^m,\ m \to \infty$) one obtains a density-1 sequence that satisfies the claim of theorem 2 for the function a. Using a diagonal argument it is possible to find a sequence equidistributed in phase space with density 1 for every function a, which closes the proof we were looking for.
From now on we will consider WLOG L_a=0 (translating the observables $a \to a - L_a$), so that the theorem reduces to showing that $\sum_{\lambda_j \in [R,2R]} |V_j(a)|^2 \to 0$ as $R \to \infty$.
Now let us try to estimate the value of $V_j(a)$ through information on the ergodic mean:

$$\langle a \rangle_T = \frac{1}{T}\int_0^T a \circ \phi_t \mathrm{d}t. \tag{3.11}$$

To do this it is necessary to find the link between the semiclassical limit of the quantum evolution and the Hamiltonian flow. We denote by $U(t) = U(t,h)$ the propagator of the free particle on the manifold.

$$U(t) = \exp\left(ith\nabla_g^2/2\right) : L^2(M) \to L^2(M). \tag{3.12}$$

$\forall a \in C_c^\infty(T^*M)$ we trivially have:

$$V_j(a) = \langle Op_{h_j}(a)u_j, u_j\rangle = \langle U(-t,h_j)Op_{h_j}(a)U(t,h_j)u_j, u_j\rangle \tag{3.13}$$

The semiclassical limit of this time evolution is given by *Egorov's theorem*:

Lemma 3.8: Egorov

$$U(-t,h_j)Op_{h_j}(a)U(t,h_j) = Op(a\circ\phi_t) + \ O_t(h_j).$$

Where ϕ_t is the Hamiltonian flow of the free particle on M, equivalent to the geodesic flow. The theorem can be obtained by finding the error made by replacing the commutators with the Poisson brackets ($\mathrm{O}(h^2)$), in the case of time evolution. Combining the last two results one obtains:

$$V_j(a) = V_j(a\circ\phi_t) + \ O_t(h_j). \tag{3.14}$$

Since V depends linearly on a, it is possible to take the ergodic mean:

$$V_j(a) = V_j(\langle a\rangle_T) + \ O_T(h_j). \tag{3.15}$$

Having assumed that $\int_{S^*M} a \ \mathrm{d}\mu_L$. the ergodic theorem establishes that for ergodic flows:

$$\lim_{T\to\infty} \|\langle a\rangle_T\|_{L^2(S^*M,\mathrm{d}\mu_L)} = 0. \tag{3.16}$$

Since V_j depends on $\langle a\rangle_T$ we can hope that the smallness of the latter term imposes a bound on V. Indeed this turns out to be true.

Lemma 3.9: Local Weyl bound

There exists a constant C such that $\forall a \in C_c^\infty(T^*M)$ and $R \geq 1$ one obtains:

$$R^{-n} \sum_{\lambda_j\in[R,2R]} |V_j(a)|^2 \leq C\|a\|^2_{L^2(S^*M,\mathrm{d}\mu_L)} + \ O_a(R^{-1})$$

.

Before showing the proof of the lemma, let us see how it closes the proof of the theorem.

$$R^{-n} \sum_{\lambda_j\in[R,2R]} |V_j(a)|^2 = R^{-n} \sum_{\lambda_j\in[R,2R]} |V_j(\langle a\rangle_T)|^2 + \ O_T(R^{-1}) \leq C\|\langle a\rangle_T\|^2_{L^2(S^*M,\mathrm{d}\mu_L)} + \ O_T(R^{-1})$$

Where we have used both the local bound and Weyl's law to simplify the sum. We observe that the term on the left does not depend on T„ while the one on the right depends on R only in the error $O(R^{-1})$. Taking the limit both for $R\to\infty$ and $T\to\infty$ one obtains:

$$\lim_{R\to\infty} R^{-n} \sum_{\lambda_j\in[R,2R]} |V_j(a)|^2 = 0. \tag{3.17}$$

Which closes the proof of the problem.

It remains to show the validity of the local Weyl bound. First of all we observe that, from Cauchy Schwarz, one obtains:

$$|V_j(a)| = \left|\langle Op_{h_j}(a)u_j, u_j\rangle\right| \leq \left\|Op_{h_j}(a)u_j\right\|_{L^2}$$

. Defining a smooth function χ that approximates the indicator function of $[1,2]$ and using the semiclassical identity $Op_h(a)^* Op_h(a) = Op_h(|a|^2) + \ O(h)+$ Weyl's law one obtains:

$$R^{-n} \sum_{\lambda_j \in [R,2R]} |V_j(a)|^2 \leq R^{-n} \sum_j \chi(\frac{\lambda_j^2}{R^2}) \left\| Op_{h_j}(a) u_j \right\|_{L^2}^2 \tag{3.18}$$

$$= R^{-n} \sum_j \chi(\frac{\lambda_j^2}{R^2}) \langle Op_{h_j}(|a|^2) u_j, u_j \rangle + \ O(R^{-1}). \tag{3.19}$$

The term on the right looks like a trace, were it not for the j-dependence of $Op_{h_j}(|a|^2)$. To remove this dependence, we rescale the semiclassical parameter.

$$h := \frac{1}{R}, \quad \tau_j := \frac{h}{h_j} = \frac{\lambda_j}{R}. \tag{3.20}$$

We define the dilation of a function on T^*M b on the momentum coordinates:

$$Op_h(b) = Op_{h_j}(\Lambda_{\tau_j} b), \quad \Lambda_\tau b(x,\xi) := b(x, \tau\xi). \tag{3.21}$$

So that $V_j(\Lambda_{\tau_j} b) = \langle Op_h(b) u_j, u_j \rangle$.
Now we fix b such that $\Lambda_\tau b = |a|^2$ on S^*M for $\tau \in supp(\chi)$, for example by setting $b(x, r\theta) = \psi(r)|a(x,\theta)|^2 \ \forall (x,\theta) \in S^*M$ and $\psi \in C_c^\infty((0,\infty))$ a cutoff function equal to 1 for $r \in (1,2)$. From the semiclassical condition of tangency on the sphere $\left\| Op_{h_j}(|a|^2 - \Lambda_{\tau_j} b) u_j \right\|_{L^2} = O(R^{-1})$ one obtains the loss of the dependence on h_j:

$$R^{-n} \sum_{\lambda_j \in [R,2R]} |V_j(a)|^2 \leq R^{-n} \sum_j \chi(\frac{\lambda_j^2}{R^2}) \langle Op_{h_j}(b) u_j, u_j \rangle + \ O(R^{-1}) \tag{3.22}$$

$$= R^{-n} tr(\chi(-h^2 \nabla_g^2) Op_h(b)) + O(R^{-1}) \tag{3.23}$$

From the semiclassical product $\chi(-h^2\nabla_g^2) Op_h(b) = Op_h(\chi(|\xi|_g^2) b) + O(h)$, and from the trace formula one has:

$$R^{-n} \sum_{\lambda_j \in [R,2R]} |V_j(a)|^2 \leq (2\pi)^{-n} \int_{T^*M} \chi(|\xi|_g^2) b(x,\xi) \mathrm{d}\xi \mathrm{d}x + O_a(R^-1) \tag{3.24}$$

$$\leq C \|a\|^2_{L^2(S^*M, \mathrm{d}\mu_L)} + O_a(R^{-1}). \tag{3.25}$$

Which closes the proof.

3.4 Examples

Let us see some examples of compact manifolds without boundary and how they relate to the hypotheses of the problem. The simplest example of a manifold of this type is the circle. Its geodesic flow trivially produces a uniform circular motion, and from periodicity it is clear that for certain times (multiples of the period) the flow cannot be ergodic in S^*S^1. Looking at the eigenvalue problem of the Laplacian, the sequence is equidistributed in physical space (from $|u_k|^2 = 1$ with $u_k = e^{ihx}$) but not in phase space (in particular there is a concentration in the direction $\xi = 1$, testable with the function $a(x,\xi) = \xi$). To find examples of ergodic systems it is necessary to study the behaviour of the geodesics, in particular their tendency to move apart. To do this we need two more notions of Riemannian geometry. The first, for 2D manifolds, is

the Gaussian curvature. Given a pair of linearly independent vectors u, v in the tangent space T_pM, which span the section σ, one defines the Gaussian curvature K:

$$\frac{\langle R(u,v)v, u\rangle}{\langle u,u\rangle\langle v,v\rangle - \langle u,v\rangle^2}. \tag{3.26}$$

The definition can be generalised to generic manifolds by considering as section a two-dimensional section and a pair of vectors that generate it. In coordinates:

$$K_{ij} = \frac{R_{ijij}}{g_{ii}g_{jj} - g_{ij}^2} = \frac{R_{ijij}}{det(g)}. \tag{3.27}$$

Where the indices are not summed. Note that in the case $dim(M) = 2$, the Riemann tensor, the Ricci tensor and the scalar curvature can be rewritten as:

$$R_{ijkl} = K(g_{ik}g_{jl} - g_{il}g_{jk}), \quad Ric_{ij} = Kg_{ij}, \quad R = 2K. \tag{3.28}$$

The second step consists in showing how the Gaussian curvature is linked to the spreading apart of geodesics. Given a geodesic $\gamma_s(t)$ we define a smooth one-parameter family of geodesics $\gamma_s(t)$ such that $\gamma_0(t) = \gamma(t)$. The *Jacobi field* J(t) is defined as:

$$J(t) = \partial_s \gamma_s(t)_{|s=0}. \tag{3.29}$$

We say that the Jacobi field is *transversal* if $g(\dot{\gamma(t)}, J(t)) = 0 \quad \forall t$. In general the field can always be decomposed into a transversal component and a parallel one. Starting from the geodesic equation one sees that $J(t)$ satisfies the Jacobi equation:

$$\frac{D^2J}{dt^2} + R(J, \dot{\gamma})\dot{\gamma} = 0. \tag{3.30}$$

Where $\frac{DJ}{dt} = \dot{X}^i \boldsymbol{\nabla}_i$, where $\boldsymbol{\nabla}_i$ is the covariant derivative with the Levi Civita connection. In a 2 D space the behaviour of $J(t)$ is entirely determined by the Gaussian curvature. We observe that if we write $J(t) = j(t)E(t)$, where $E(t)$ is a field transversal to the geodesic, i.e. such that it satisfies $\frac{D}{dt}E(t) = 0$, the equation above reduces to:

$$j''(t) + K(\gamma(t))\, j(t) = 0. \tag{3.31}$$

Where this time the symbol "'" is to be understood as an ordinary derivative.

An example to put into practice the definitions seen above is obtained by considering the complex upper half-plane $\mathcal{H}$, with the hyperbolic metric:

$$\mathrm{d}s^2 = \frac{\mathrm{d}x^2 + \mathrm{d}y^2}{y^2}. \tag{3.32}$$

From the explicit computation one checks that the curvature is constant and equals $K = -1$. The differential equation for the Jacobi field becomes:

$$j''(t) - j(t) = 0. \tag{3.33}$$

Which has explicit solution $j(t) = Ae^{-t} + Be^{t}$. This equation is uniquely determined by imposing the initial conditions $j(0), j'(0)$. For example for $j(0) = 0,\ j'(0) = v$ one has $j(t) = v\sinh(t)$. We

observe that generically $j(t)$ diverges exponentially from the original value. More precisely, one can define $J_t(x,v) = D\phi^t_{x,v}(w)$, where D is the differential of the geodesic flow. Consequently

$$D\phi^t : TSM \to TSM, \quad ((x,v),w) \to (\phi^t(x,v), D\phi^t(w)). \tag{3.34}$$

The previous example shows how a negative curvature (-1) implies the hyperbolic spreading apart of the geodesics. The formalisation of this idea, and of how it leads to ergodicity, is given by Hopf's theorem:

Theorem 3.10: Hopf

Let (M,g) be a compact Riemannian manifold, C^2 and with strictly negative sectional curvature. Then:

1. The geodesic flow is Anosov,
2. the Liouville measure is invariant under the geodesic flow;
3. the system (SM, ϕ^t, μ_L) is ergodic.

Going back to the example of hyperbolic space, it seems a good candidate to exhibit ergodicity. The only problem that at the moment does not allow us to satisfy the previous theorem is that, as a manifold, it is not compact. Let us therefore see how it is possible to compactify the space. The isometries of $\mathbb{H}$ are classified by the Möbius transformations:

$$gz = \frac{az-b}{cz-d}, \quad a,b,c,d \in \mathbb{R}. \tag{3.35}$$

In group terms, these are isomorphic to an action of the matrix group $PSL(2,\mathbb{R})$. An important property of this action is that it is transitive. In particular it is possible to write $\mathbb{H}$ as a homogeneous space:

$$\mathbb{H} \cong SL(2,\mathbb{R})/SO(2). \tag{3.36}$$

An important consequence is that it is possible to write a generic element of SL through the Iwasawa decomposition:

$$\begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix} \begin{pmatrix} y^{1/2} & 0 \\ 0 & y^{-1/2} \end{pmatrix} \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}. \tag{3.37}$$

Following the isomorphism of the upper half-plane, it is easy to see that the unit bundle is homeomorphic to the Möbius group:

$$S\mathbb{H} \cong PSL(2,\mathbb{R}). \tag{3.38}$$

To compute the geodesic flow, we observe that every geodesic can be obtained from the geodesic geometrically defined by $Re(z) = 0$ through the application of an isometry. Focusing therefore only on the imaginary line, knowing that the distance $d(i,\alpha i) = \log(\alpha)$, the geodesic flow starting from the point (i,i) can be written as:

$$a_t(i,i) = \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix} (i,i). \tag{3.39}$$

In general, therefore

$$a_t(x,y) = a_t g \circ (i,i) = g \circ \begin{pmatrix} e^{t/2} & 0 \\ 0 & e^{-t/2} \end{pmatrix} (i,i). \tag{3.40}$$

Finally, it is useful to know that $PSL(2,\mathbb{R})$ has 3 conjugacy classes, depending on whether the modulus of the trace of the matrix $|Tr(\gamma)|$ is $>$ (hyperbolic matrices), $=$ (parabolic) or $<$ than 2 (elliptic).
To obtain the compact manifold we are looking for, what one does is to take the quotient $\Gamma \setminus \mathbb{H}$, where Γ is a discrete subgroup of $SL(2,\mathbb{R})$. In general Γ cannot be taken in a totally arbitrary way, but it must satisfy the following conditions:

1. To be a Riemannian manifold, Γ must not contain any elliptic element. Otherwise it can have singularities, and turn out to be the more general structure of an *Orbifold*.

2. Γ must be cocompact, i.e. there must exist a compact set $K \in \mathbb{H}$ such that $\Gamma \circ K = \mathbb{H}$. Under this condition, the quotient is compact.

An important example of this quotient, even if not compact, is given by modular forms, in which $\Gamma = PSL(2,\mathbb{Z})$. In the space $PSL(2,\mathbb{Z}) \setminus \mathbb{H}$, "infinity" is characterised by a cusp, which gives the space finite volume.

4. Some extensions of the theorem

Given the formulation of the quantum ergodic theorem, it is legitimate to ask which extensions are possible. In this section, such extensions are proposed only in the form of questions, without any further investigation.

- We have seen that under suitable hypotheses the theorem ensures the existence of subsequences of eigenfunctions with density 1, such that:

$$\langle Op_{h_{j_k}}(a) u_{j_k}, u_{j_k} \rangle \to \int_{S^*M} a(x,\xi) \mathrm{d}\mu_L(x,\xi). \tag{4.1}$$

  We call *semiclassical measure* any other kind of measure $\mathrm{d}\mu$ to which a subsequence can converge. At this point it is legitimate to ask: given the sequence of eigenfunctions, is there only one semiclassical measure?

- In the theorem the manifold is assumed to be compact. Is it possible to weaken this hypothesis by requiring only that the volume be finite, as in the case of modular forms?

- In classical dynamical systems the Sinai-Kolmogorov entropy plays an important role in determining various properties of the motion. What information does it provide to semiclassical measures?

5. References

- "Around quantum ergodicity" by Semyon Dyatlov, for the review of the proof of the quantum ergodic theorem, and of the possible extensions;
- "Quantum chaos on hyperbolic surfaces" by Etienne Le Masson, for the treatment of the geodesics for hyperbolic surfaces.